

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • Stamp 1G — Full-Time Work Eligible

PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Developed and validated **R-Shiny programs** for statistical analysis and reporting of clinical trial data, supporting the interpretation of results for **data science** initiatives.
- Collaborated with biostatisticians and cross-functional teams to interpret study requirements, ensuring accurate implementation of statistical analyses for **clinical trial data**.
- Prepared and reviewed **statistical analysis plans (SAPs)** and programming specifications, implementing derived datasets to meet **regulatory standards** for clinical research.
- Ensured data integrity and compliance with **ICH/GCP guidelines** throughout the programming process, maintaining audit trails for critical research data.
- Mentored junior programmers on best practices in **R programming** and statistical analysis techniques, contributing to team skill development.
- Automated data extraction and transformation processes using **Python scripting**, reducing manual effort by an estimated **30%** for recurring reporting tasks.

SKILLS

Programming Languages – R, R-Shiny, SAS, Python

Data Analysis & Statistics – Statistical Analysis, Data Modelling, Exploratory Data Analysis, Hypothesis Testing, Time-Series Analysis, Machine Learning Fundamentals

Clinical Research Standards – ICH/GCP Guidelines, Regulatory Standards, Clinical Trial Analysis

Data Management & ETL – Clinical Data Management Systems, Power Query, SQL (PostgreSQL, MySQL, Oracle)

BI & Visualization – Power BI, Tableau, DAX Measures, R-Shiny

Development Tools & Practices – Git, Jenkins, Jira, Confluence, Agile/Scrum, REST APIs, Docker

PROJECTS

Clinical Trial Analysis Dashboard (HCL Technologies)

- Developed an interactive **R-Shiny dashboard** to visualize key performance indicators from clinical trial datasets, enabling faster insights for operational teams.
- Implemented data transformation logic using **R and SQL** to aggregate and prepare data for analysis, handling approximately **50,000 patient records** per trial.
- Collaborated with business analysts to define reporting requirements and ensure the dashboard met user needs for monitoring trial progress and outcomes.

Automated Reporting Pipeline (HCL Technologies)

- Engineered a **Python-based ETL pipeline** to automate the generation of weekly operational reports from multiple enterprise systems.
- Integrated with **PostgreSQL and Oracle databases** to extract and validate data, reducing report generation time from 3 hours to under 45 minutes.
- Deployed the solution using **AWS Lambda and CloudWatch** for scheduled execution and monitoring, ensuring reliable delivery of critical business intelligence.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)